Credit Card Fraud Detection Using Machine Learning

Internship Project Report | Elevate Labs

Abstract

This project demonstrates a machine-learning workflow for identifying potentially fraudulent credit-card transactions. It combines two unsupervised anomaly-detection methods—Isolation Forest and Local Outlier Factor (LOF)—with a supervised Random Forest classifier. A Streamlit dashboard presents dataset information, model comparison metrics, a confusion matrix, an ROC curve, and an interactive prediction form. The app supports uploading a labelled CSV dataset with a target column named **Class**, where 0 represents legitimate transactions and 1 represents fraud.

Introduction

Card-payment fraud can cause financial losses for customers and businesses. Machine learning can help flag suspicious patterns for further review. Because fraud cases are usually much less common than legitimate transactions, evaluation should consider precision, recall, F1-score, and ROC-AUC rather than accuracy alone. This project is an educational prototype and is not intended to make real banking decisions.

Tools Used

- Python
- Pandas and NumPy for data handling
- Scikit-learn for preprocessing, anomaly detection, classification, and evaluation
- Matplotlib for charts
- Streamlit for the interactive web dashboard
- GitHub for version control and project hosting

Steps Involved in Building the Project

1. Load a labelled CSV file or use the built-in synthetic demo dataset to check the interface.
2. Clean missing and infinite values, select numeric features, and separate the target column.
3. Split the dataset into training and testing sets using stratification; standardize features using the training set.
4. Fit Isolation Forest and Local Outlier Factor to flag anomalous transactions.
5. Train a class-weighted Random Forest classifier and produce fraud probability scores.
6. Evaluate predictions using precision, recall, F1-score, a confusion matrix, and the ROC curve / ROC-AUC.
7. Display results in Streamlit and allow a user to enter feature values for a demonstration prediction.

Current Status and Evaluation

The Streamlit dashboard has been run locally and the source files app.py, README.md, and requirements.txt have been uploaded to GitHub. The current screenshot shows the application running in demo mode with 5,000 synthetic records, including 125 generated fraud-labelled records. These synthetic-data metrics are only a software demonstration and must not be presented as validated real-world performance. For final dataset-based evaluation, upload the Kaggle Credit Card Fraud Detection dataset (commonly named creditcard.csv) and rerun the app. The app uses Random Forest as its supervised baseline; XGBoost is not currently implemented.

Conclusion

The project provides an end-to-end demonstration of data preprocessing, anomaly detection, supervised classification, evaluation, and interactive deployment with Streamlit. A next improvement is to evaluate on the real labelled dataset, tune the decision threshold based on the precision–recall trade-off, and add XGBoost if required by the project brief. The model should only be used as an educational prototype until it is tested and validated on suitable real data.